

SMART AI-POWERED SERVICE DESK

System Design & Technical Documentation

1. System Overview

The Smart AI-Powered Service Desk combines FAQ Search, RAG-based Knowledge Search, GPT-powered Ticket Classification, Agent Queue Management, and Admin Governance. The platform reduces manual effort by automatically classifying tickets into IT, HR, and Facilities categories and assigning priorities.

2. Customer Workflow

User → FAQ Search / RAG Search → Answer Found? → Display Answer. If no answer is found, the user logs in, creates a ticket, and GPT classifies the request.

3. RAG Workflow

Query Ingestion → Embeddings (Sentence Transformers) → ChromaDB Search → Retrieve Relevant Chunks → GPT Response Generation → User Answer.

4. Ticket Classification Workflow

Ticket Subject + Description → GPT Analysis → Department Classification (IT/HR/Facilities) → Priority Assignment (High/Medium/Low) → Save Ticket.

5. Agent Workflow

Agent Login → Dashboard → Queue → Open Ticket → View Conversation → Reply → Update Status → Close Ticket.

6. Admin Workflow

Admin uploads PDFs/DOCX files → Chunking → Embedding Generation → ChromaDB Storage → Available for Semantic Search.

7. System Architecture

Frontend: React + Tailwind CSS Backend: FastAPI Database: PostgreSQL Vector DB: ChromaDB AI Layer: GPT-5.2 + Sentence Transformers

8. Database Design

Users, Agents, Tickets, Ticket Messages, FAQs, Knowledge Base Documents.

9. API Overview

Authentication APIs, Ticket APIs, Agent APIs, Admin APIs, RAG Search APIs.

10. Non-Functional Requirements

Security: JWT Authentication Performance: RAG < 5 seconds Scalability: Docker Deployment Availability: 99% uptime

ASCII Workflow Diagrams

USER QUERY | v EMBEDDINGS | v CHROMADB | +---+---+ | | FOUND NOT FOUND | | GPT
CONTACT AGENT | ANSWER